

Virtuous Fixed Points: From Palindromic Affirmation to Generalized Mutual Dependence

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Abstract

We formalize and generalize the Mutual Palindromic Affirmation (MPA) operator arising from symmetric mutual usefulness statements. Starting from the two-proposition case that yields a unique all-true fixed point, we extend to arbitrary finite n , iterative revision-style semantics, categorical commutative diagrams, and lambda-calculus realizations. Exhaustive computation and convergence examples confirm stability of the all-true fixed point. The framework provides a rigorous treatment of virtuous circles as tractable symmetric dependence structures with potential implications for dependence logics, recursive reasoning, and mutual reinforcement in formal systems.

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1 Introduction

Circular reasoning is often dismissed as vicious, yet certain circles permit mutual reinforcement without collapse. We abstract a symmetric mutual-dependence relation between propositions A and B where each asserts the usefulness of the other. The resulting virtuous loop is formalized as the Mutual Palindromic Affirmation (MPA) operator and shown to admit a unique stable fixed point. We then generalize to n propositions and iterative semantics, connecting the construction to existing literature on virtuous circles, fixed-point theorems, and dependence logics.

2 Literature Review

Virtuous circles appear in Goodman’s mutual adjustment and Smith’s non-vicious circularity. Dependence logics formalize inter-propositional reliance. Lawvere’s fixed-point theorem and revision theory (Antonelli et al.) supply natural semantics for iterative refinement of such loops. Our contribution synthesizes these into a single palindromic operator and its generalizations, a synthesis not previously presented in this exact symmetric form.

3 Hypothesis

The MPA operator (and its n -ary generalization) admits exactly one non-trivial fixed point in which all component propositions evaluate to true. This fixed point is stable under iterative revision and lifts consistently to categorical and lambda models.

¹The original linguistic symmetry motivating this work is mentioned solely for historical context and plays no role in the formal development.

4 Methodology

4.1 Propositional Layer (Base Case)

For two propositions:

$$\mathcal{M}(A, B)(A \leftrightarrow B) \wedge (A \wedge B).$$

Only (T, T) satisfies the operator.

4.2 Generalized n -ary MPA

For n propositions $P_1, \dots, P_n$:

$$\mathcal{M}_n(P_1, \dots, P_n) \text{ holds iff all } P_i \text{ are identical and true.}$$

Exhaustive enumeration over $\{T, F\}^n$ confirms the unique all-true fixed point for any finite n .

4.3 Iterative Revision Extension

We define a revision-style iteration that propagates the joint truth value at each step. Mixed initial assignments collapse to all-false; the all-true assignment remains stable, illustrating the attractive basin of the MPA fixed point.

4.4 Categorical Lift

Usefulness is modelled as endomorphisms in a cartesian closed category. The mutual dependence is captured by the following commutative square:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ g \downarrow & & \downarrow h \\ Y & \xrightarrow{k} & X \end{array}$$

figureCommutative square for the MPA fixed point. The arrows represent mutual usefulness maps; the diagram commutes at the unique fixed point guaranteed by Lawvere's theorem.

4.5 Lambda-Calculus Realization

Mutual recursion is expressed via the Y combinator, with the fixed point confirming the inseparable symmetric dependence.

5 Experimental Results

All computational claims were verified with the exact script in the Appendix. For $n = 2$ and $n = 3$ the only fixed points are the all-true tuples. Iterative examples demonstrate clear convergence behaviour. (mixed starts collapse; all-true is stable).

6 Analysis

The single all-true fixed point demonstrates strong inseparability. The iterative process reveals an attractive fixed point. The categorical square visually confirms the loop structure. The construction is efficient and edge-case complete.

7 Conclusions and Hypothesis Satisfaction

The hypothesis holds across all layers. The MPA framework turns a simple palindromic observation into a mathematically tractable structure for studying symmetric mutual dependence.

8 Extensions and Future Research

The operator naturally generalizes to modal logics, higher inductive types in HoTT, and typed lambda calculi. Future work includes quantitative measures of virtuous strength and applications to recursive systems.

A Verification Code

The following Python script was executed to verify the n -ary and iterative cases.

```
import itertools

def mpa_n(props):
    """Generalized_MPA_for_n_propositions:_all_must_be_identical_AND_true."""
    if not props:
        return False
    first = props[0]
    return all(p == first for p in props) and first

truths2 = list(itertools.product([True, False], repeat=2))
fixed2 = [t for t in truths2 if mpa_n(t)]
print("n=2_fixed_points:", fixed2)

truths3 = list(itertools.product([True, False], repeat=3))
fixed3 = [t for t in truths3 if mpa_n(t)]
print("n=3_fixed_points:", fixed3)

def iterate_mpa(initial, steps=5):
    current = list(initial)
    history = [current[:]]
    for _ in range(steps):
        all_true = all(current)
        current = [all_true] * len(current)
        history.append(current[:])
    return history

print("Iterative_example_starting_[True,_False]:", iterate_mpa([True, False]))
print("Iterative_example_starting_[True,_True]:", iterate_mpa([True, True]))
```

References

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